

Principles of Data Visualization

Decoding

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Outline

Introduction

Principles of Data Visualisation

Scale and range

Theories of perception

Decoding: William Cleveland

A hierarchy of visual decoding

Size (numeric)

Group (categorical)

Colour

Aesthetics: Tufte

Final words

Contents of this course

How to make an informative visual representation of data and/or results

- Part I: process of graph construction (encoding)
 - How to represent (“map”) data/results in graph
 - The structure of a graph; the grammar of graphics
- Part II: principles of graph perception and interpretation (decoding)
 - What happens when you see and read a graph
 - Principles for creating truthful and informative graphs

Points to consider

- **Why** do we make a graph? (purpose)
 1. We want to explore and inspect **our** data or results
 2. Present data or results, often with a message, to **others**
- **Where** do we show the graph: research paper, conference, website
- **Who** is the audience. Level of background knowledge
- **What** do we want to show (data)
 - Graph: visual representation of information (“encoding”) mapping from data/results to figure
 - Show values in isolation and/or focus on patterns

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Grammar of Graphics (R terminology)

Statistical Graph mapping from data to aesthetic attributes of geometric objects

Geometric Object what you see on the plot (points, lines, bars)

Aesthetic Attributes “role” of variables in plot: location (often along x-axis and y-axis), colour, size, shape

Scales map data values to aesthetic attributes.

Maybe with position adjustment along axis (stacked or dodged barplot, point jitter)

Statistical Transformation

Coordinate System Cartesian, polar, map projections.
(Networks, trees have no coordinate system)

Faceting separate panels (subfigures)

Grammar of Graphics (R terminology)

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Statistical Transformation

Coordinate System Cartesian, polar, map projections.
(Networks, trees have no coordinate system)

Faceting separate panels (subfigures)

Not about **graph type**; ($\approx$ language, words).

Not about font size, background colour . . . Can be specified in a **theme**

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Scales on the axes, some principles

- Linear scale: values as measured
 - Most easy interpretation of values

But:

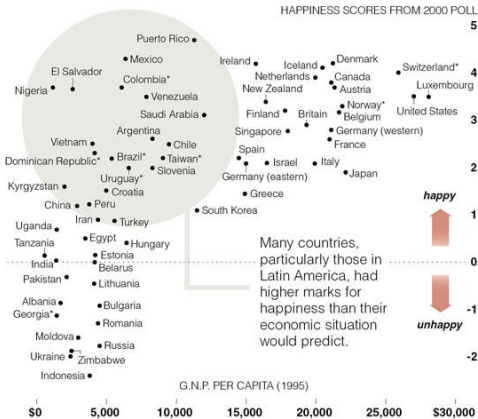
- Distribution may be very skewed
- May give wrong impression of relationship

Choice of scale influences interpretation

A Plateau of Happiness

A country's wealth may not always dictate the happiness of its people.

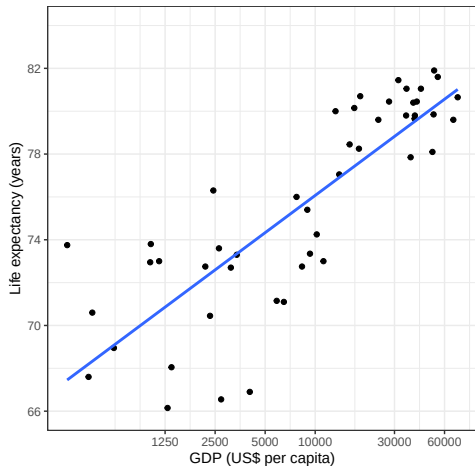
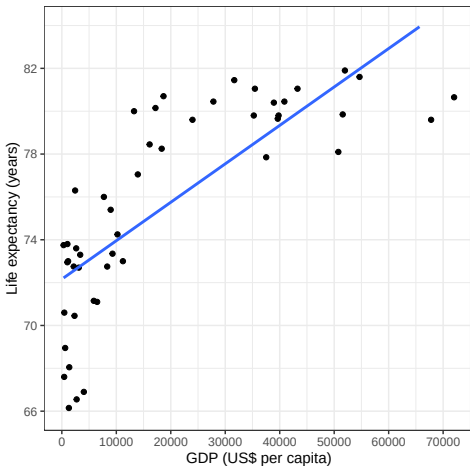
As part of the World Values Survey project, inhabitants of different countries and territories were asked how happy or satisfied they were. Below is a sampling of happiness rankings, along with economic status.



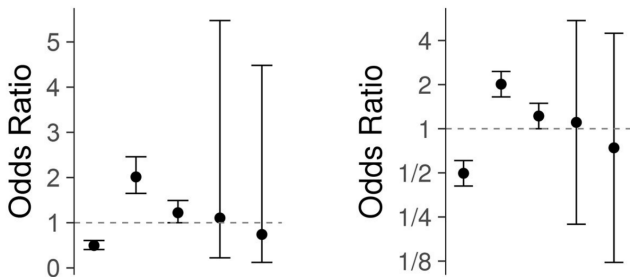
*Poll results for these countries were from 1995.

Source: Ronald Inglehart, "Human Beliefs and Values: A Cross-Cultural Sourcebook Based on the 1999-2002 Values Surveys"

WHO data 2007, two scales



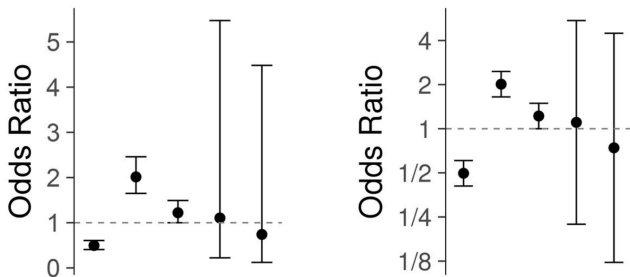
Always use a log scale for ratios



(P : probability outcome; odds = $\frac{P}{1-P}$;

$$\text{odds ratio odds}(A)/\text{odds}(B) = \frac{P_A}{1-P_A} / \frac{P_B}{1-P_B}$$

Always use a log scale for ratios



(P : probability outcome; odds = $\frac{P}{1-P}$;

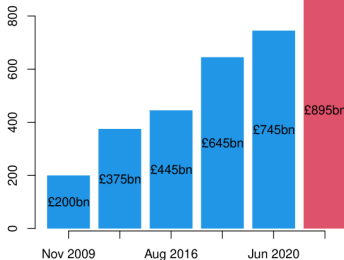
$$\text{odds ratio odds}(A)/\text{odds}(B) = \frac{P_A}{1-P_A} / \frac{P_B}{1-P_B}$$

$$\text{If } \text{OR}(F/M) = \frac{\text{odds}(F)}{\text{odds}(M)} = 2, \text{ then } \text{OR}(M/F) = \frac{\text{odds}(M)}{\text{odds}(F)} = 0.5$$

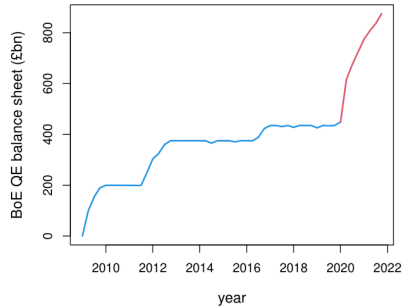
Change in Quantitative Easing

We have been increasing the amount of Quantitative Easing

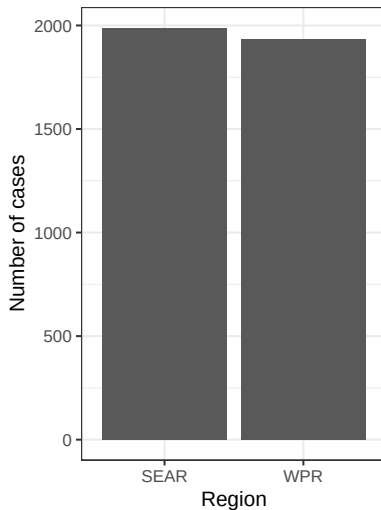
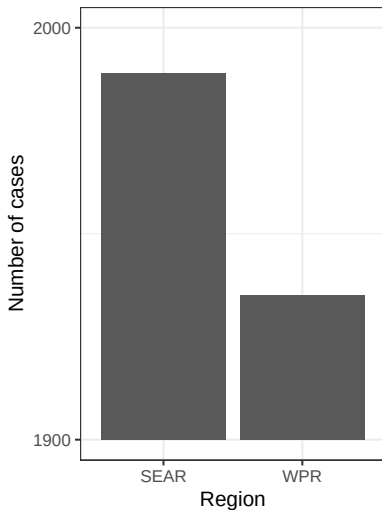
Bank of England purchases of bonds in £ billion



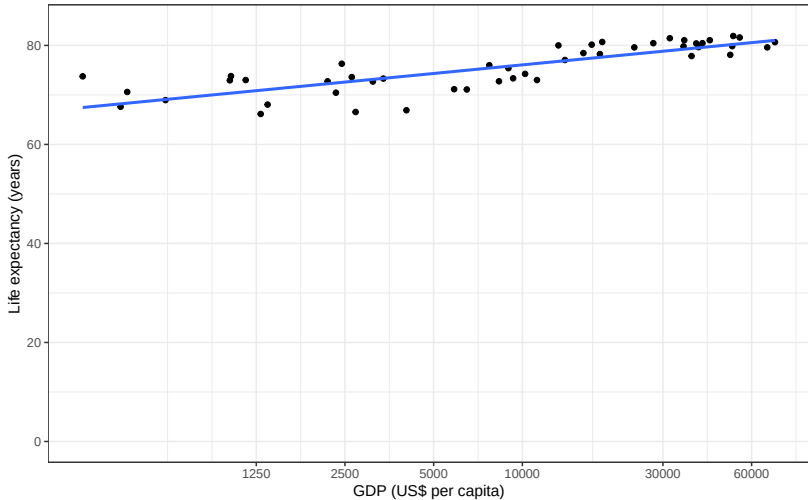
ONS quarterly data



Include zero: measles cases in October 2019, two WHO regions

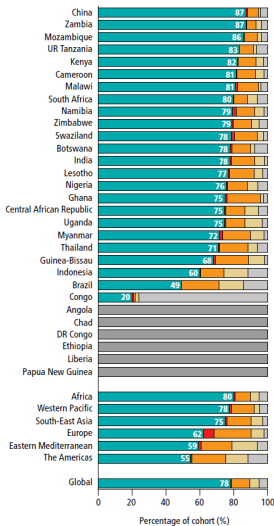


Don't include zero: no country has life expectancy < 60

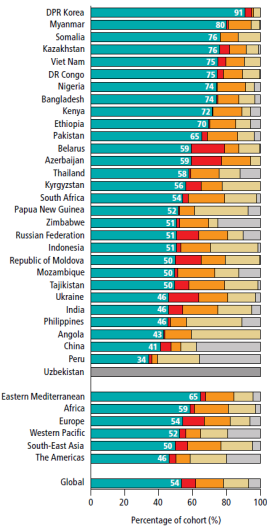


WHO Global TB Report 2017

Treatment outcomes for new and relapse HIV-positive TB cases in 2015, 30 high TB/HIV burden countries, WHO regions and globally



Treatment outcomes for rifampicin-resistant TB cases started on treatment in 2014, 30 high MDR-TB burden countries, WHO regions and globally



Criteria for graph quality

- Truthful
 - based on data collected in honest research
 - objective display of information
 - use appropriate scale and range
- Intuitive: easy to understand, concise without oversimplifying (see Section “Aesthetics: Tufte”)

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- Truthful
 - based on data collected in honest research
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 - use appropriate scale and range
- Intuitive: easy to understand, concise without oversimplifying (see Section “Aesthetics: Tufte”)
- Informative (see Section “Decoding: William Cleveland”)
 - use appropriate graph type
 - use proximity and alignment to facilitate comparison
 - use labels and annotations to add clarity to the message

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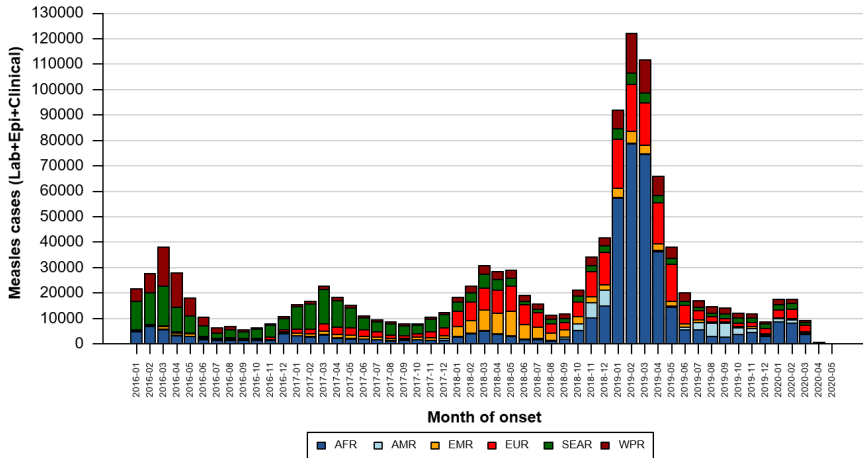
Colour

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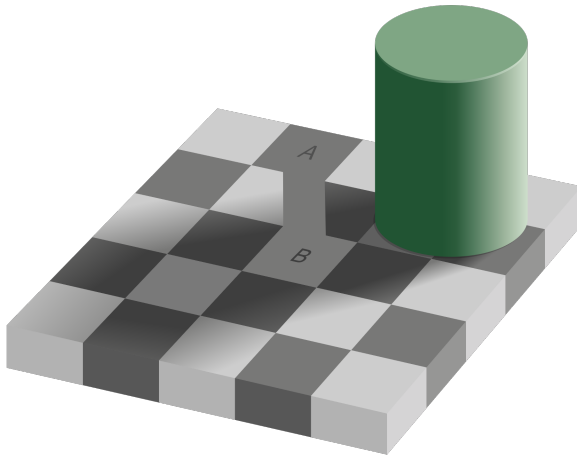
Measles case distribution by month and WHO Region (2016-2020)



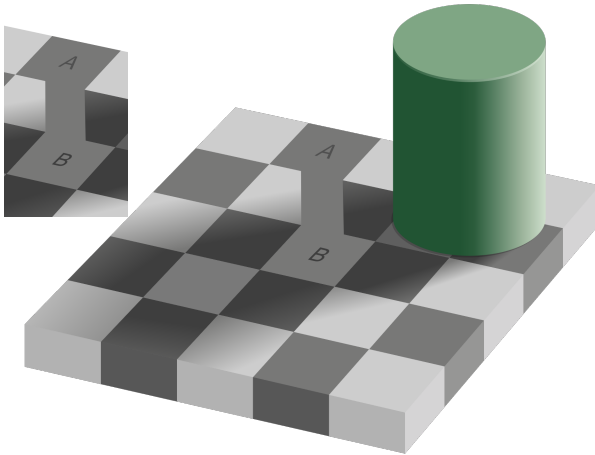
Two modes of perception: Seeing and Reading

- Daniel Kahneman: **Thinking, Fast and Slow**
 Fast: intuitive, “seeing”
 Slow: logical, “reading”
- What happens when you interpret a graph
 1. Graphical perception: first impression, **see**.
 “There was a peak in measles cases in the first half of 2019”
 2. Graphical cognition: draw conclusions, **read**.
 “The outbreak was mainly in the African region”
 “The number of cases in the South-East Asian region has declined over time”

Adelson's Checkerboard (II)



Adelson's Checkerboard (II)



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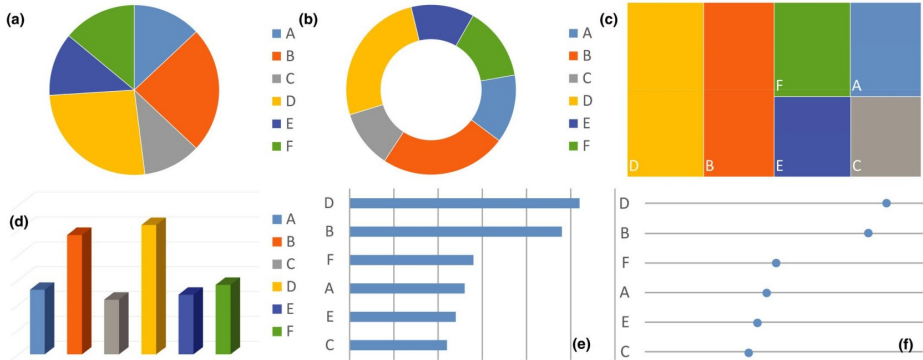
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How to visualize six numbers



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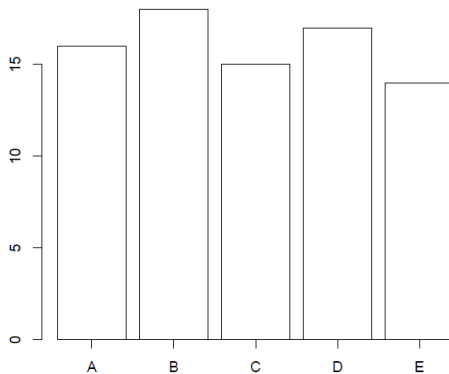
Decoding

- Table look-up: attach value to what is shown; pattern recognition
- Size (numeric)
- Group (categorical)

Decoding

- Table look-up: attach value to what is shown; pattern recognition
- Size (numeric)
 In decreasing level of efficiency:
 1. Position along common scale
- Group (categorical)

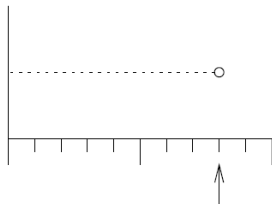
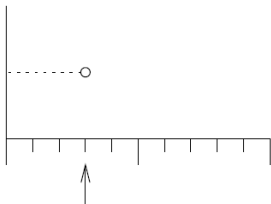
1. Position along common scale



Decoding

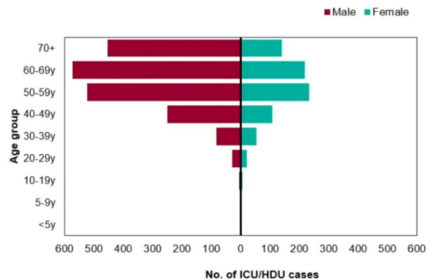
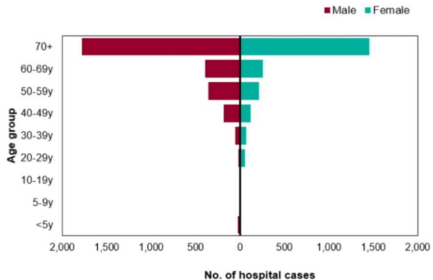
- Table look-up: attach value to what is shown; pattern recognition
- Size (numeric)
 In decreasing level of efficiency:
 1. Position along common scale
 2. Position along identical, nonaligned scales
- Group (categorical)

2. Position along identical, nonaligned scales



ICU mortality in covid-19

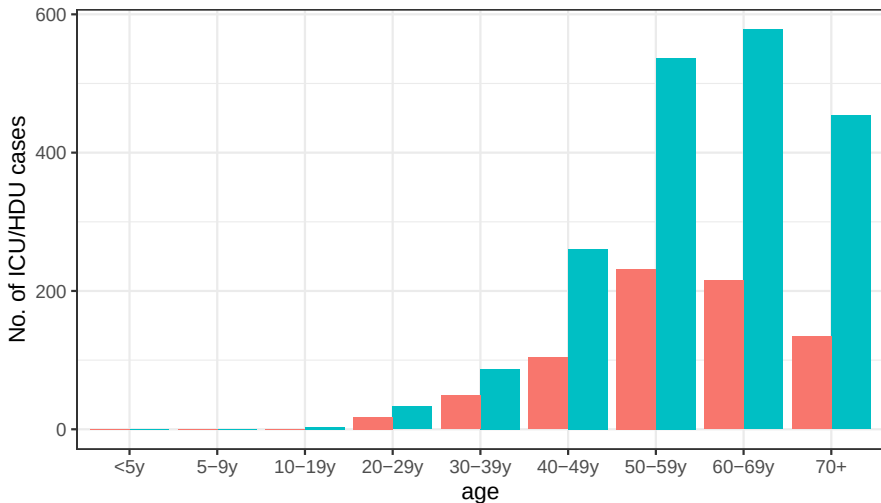
Hospitalised cases (L) and ICU/HDU cases (R).
PHE, data from England.



where men are much more
likely to have severe disease

Alternative: dodged barchart

gender Female Male

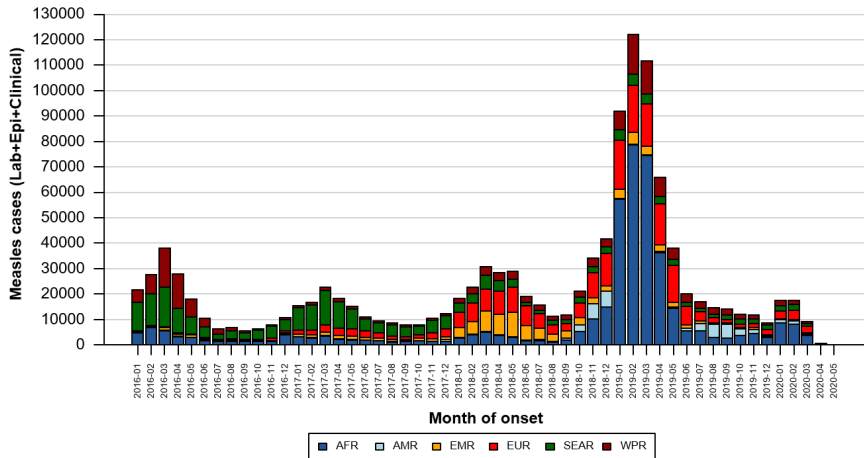


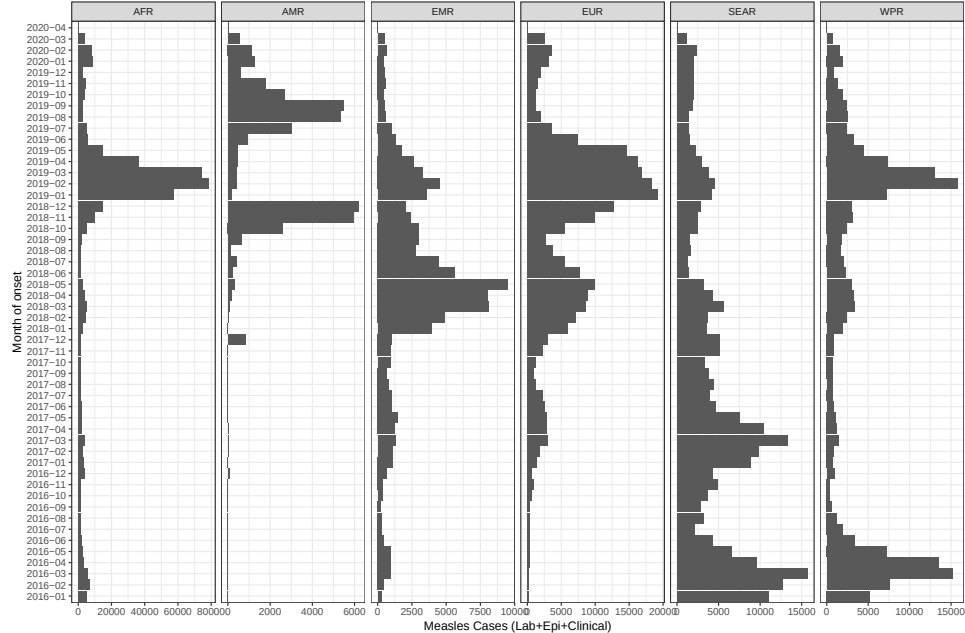
Decoding

- Table look-up: attach value to what is shown; pattern recognition
- Size (numeric)
 In decreasing level of efficiency:
 1. Position along common scale
 2. Position along identical, nonaligned scales
 3. **Length; grid lines are useful**
- Group (categorical)

3. Length

Measles case distribution by month and WHO Region (2016-2020)

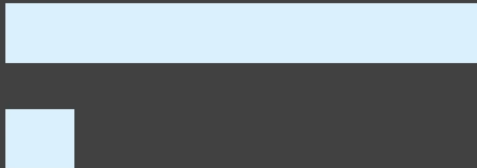




Decoding

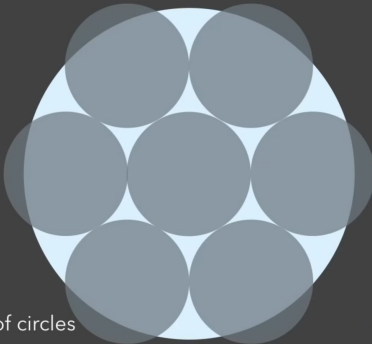
- Table look-up: attach value to what is shown; pattern recognition
- Size (numeric)
In decreasing level of efficiency:
 1. Position along common scale
 2. Position along identical, nonaligned scales
 3. Length; grid lines are useful
 4. Angle, slope; don't use pie charts
 5. Area
- Group (categorical)

How many times larger is the first bar?



Compare **length** of bars

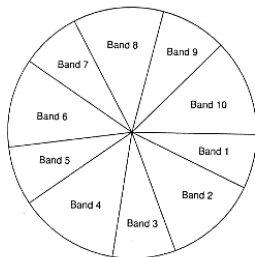




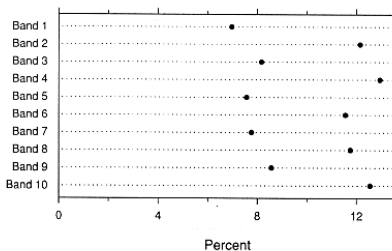
Compare **area** of circles



Angle: pie chart versus dot plot



4.19 PIE CHART. The pie chart falls in the category of a pop chart — a graphical method used frequently in the mass media and certain business presentations but far less in science and technology. Both table look-up and pattern perception are less efficient for pie charts than for dot plots.

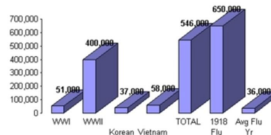


4.20 DOT PLOT. The data from Figure 4.19 are graphed by a dot plot. Patterns emerge that cannot be decoded from Figure 4.19.

Mortality determined by fatality rate and how many infected.

- Infection fatality rate of COVID-19 probably around or just below 1% (depending on age structure of country- and how many asymptomatic).
- Comparison with other diseases:
- Ebola around 70% when first emerged.
- HIV 100% when first emerged.
- Smallpox 30%.
- H1N1 2009 flu 0.1%.
- H1N1 1918-20 'Spanish' flu around 3%.

US Deaths 20th Century - Flu and War



if you combine all the people
for example in the U.S.A.

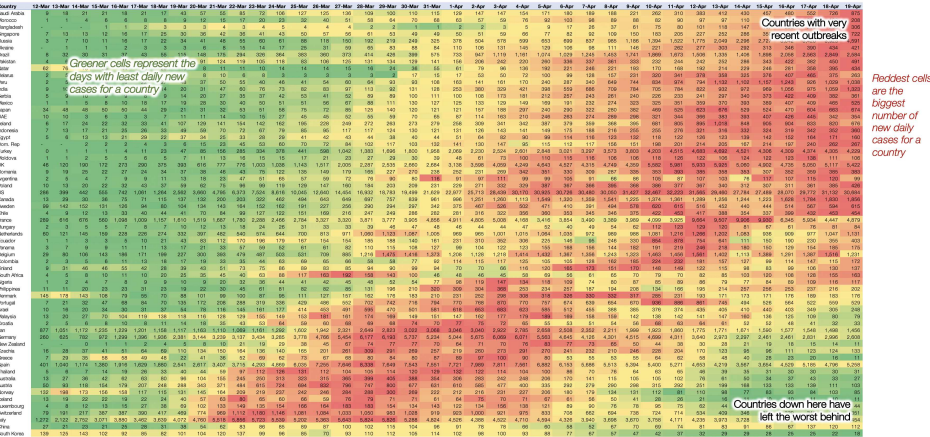
Decoding

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In decreasing level of efficiency:
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 2. Position along identical, nonaligned scales
 3. Length; grid lines are useful
 4. Angle, slope; don't use pie charts
 5. Area
 6. Volume; don't use unnecessary third dimension
 7. Colour
- Group (categorical)



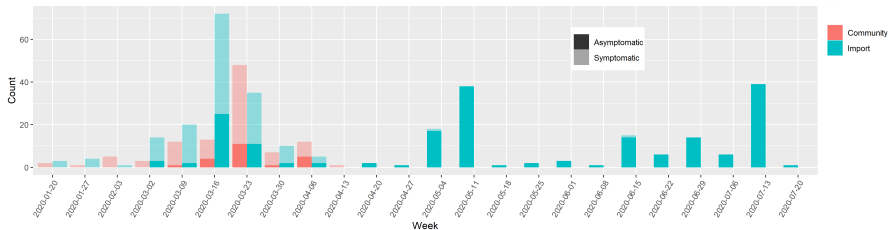
Alternative as heatmap

Chart 3: Daily New Coronavirus Cases Worldwide*

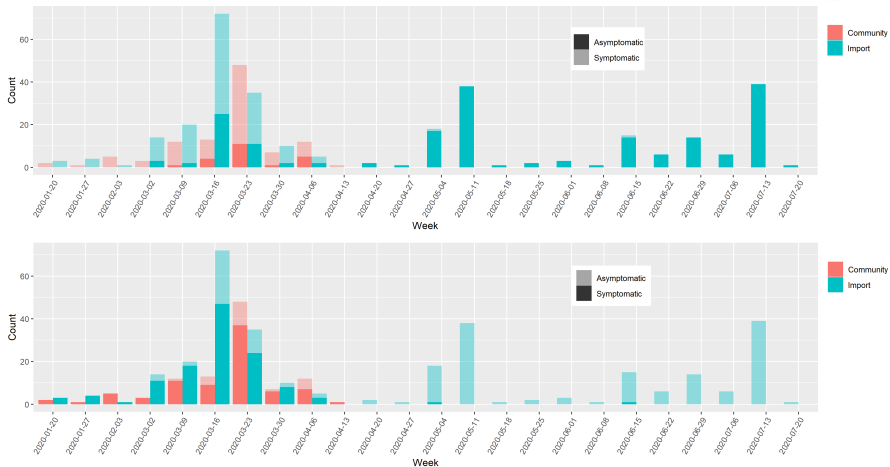


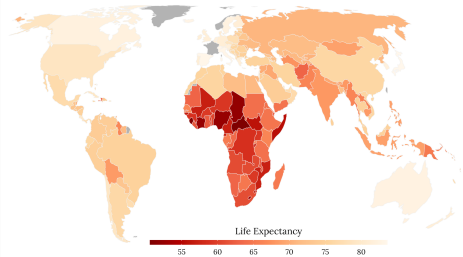
*As of April 19th 2020. Top ~60 countries in total cases, with added countries with special interest. Daily new cases calculated as 4-day averages to smooth outliers and weekend effects. Order based on average of last 5 days over average of last 25 days, adjusted manually to highlight the trend
Source: https://github.com/CSSEGISandData/COVID-19/blob/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_confirmed_global.csv

Number of SARS-CoV-2 infections in Vietnam



Number of SARS-CoV-2 infections in Vietnam





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Basics

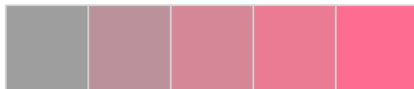
- Three colour models
 - RGB (Red-Green-Blue): how computers represent colour ($\approx$ the 3 eye cones in humans)
 - HSV (Hue-Saturation-Value):
 $\approx$ wavelength-colour purity-brightness
 - HCL (Hue-Chroma-Luminance):
 $\approx$ wavelength-colourfulness-brightness
closest to perceptual properties

HCL explained

Hue
(Type of color)



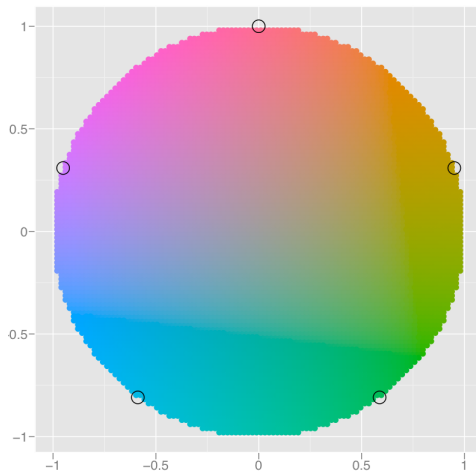
Chroma
(Colorfulness)



Luminance
(Brightness)



HCL colour wheel (fixed luminance)



Basics

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 - RGB (Red-Green-Blue): how computers represent colour ($\approx$ the 3 eye cones in humans)
 - HSV (Hue-Saturation-Value):
 $\approx$ wavelength-colour purity-brightness
 - HCL (Hue-Chroma-Luminance):
 $\approx$ wavelength-colourfulness-brightness
closest to perceptual properties
- Take colour-blindness into account:
deuteranopia: green deficient; protanopia: red deficient;
tritanopia: blue deficient

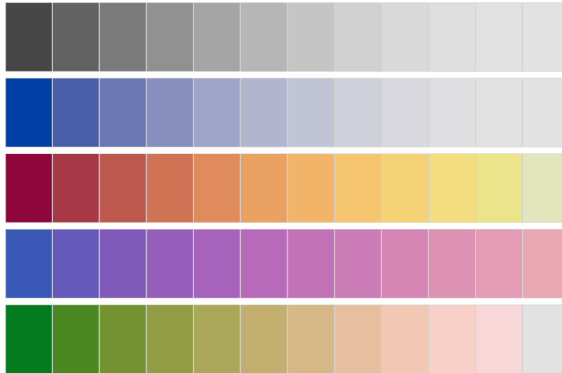
Represent value by colour: palettes

- Qualitative: categorical information
 - same perceptual weight to each category: vary hue, keep chroma and luminance fixed
 - may choose hues based on convention (red and blue for political parties in United States, cold/hot, risky/safe)

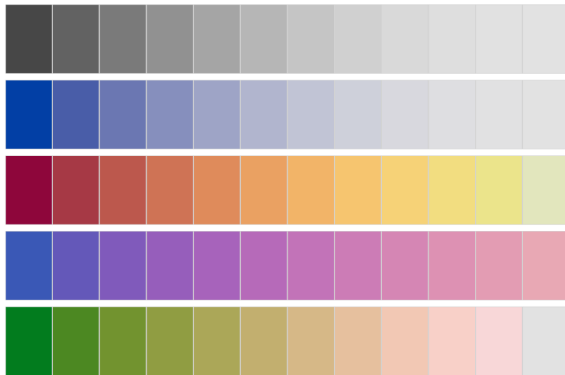
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- Sequential: for ordered/numeric information
 - Single or two hues; change luminance linearly along with value, may change chroma

Sequential palettes

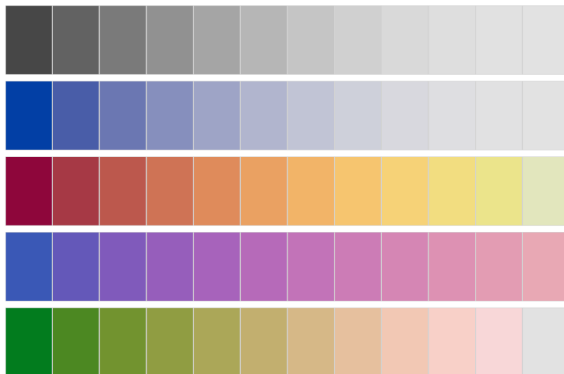


Sequential palettes



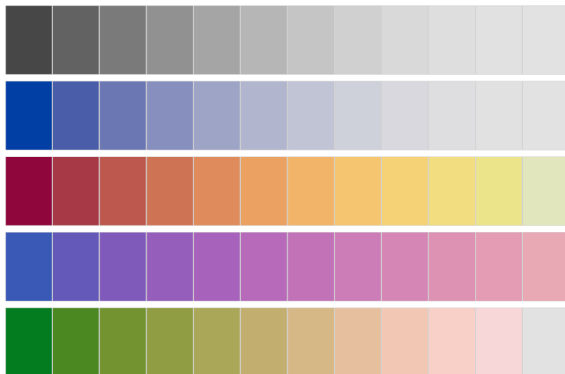
1st: vary luminance only

Sequential palettes



1st: vary luminance only; 2: vary chroma and luminance

Sequential palettes



1st: vary luminance only; 2: vary chroma and luminance;
3-5: vary hue, chroma and luminance

Represent value by colour: palettes

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 - same perceptual weight to each category: vary hue, keep chroma and luminance fixed
 - may choose hues based on convention (red and blue for political parties in United States, cold/hot, risky/safe)
- Sequential: for ordered/numeric information
 - Single or two hues; change luminance linearly along with value, may change chroma
- Diverging: for ordered/numeric information in two directions around neutral value
Combines two sequential palettes with different hues

Diverging palettes



hcl_palettes() in colorspace package

```
R> hcl_palettes(plot = TRUE)
```



Some advice on colour palettes

- Avoid large areas of flashy, highly saturated colors; use light colors (higher luminance, lower chroma). Recently changed in base R palette

Some advice on colour palettes

- Avoid large areas of flashy, highly saturated colors; use light colors (higher luminance, lower chroma). Recently changed in base R palette
- For points and lines: lower luminance, higher chroma
- If size of coloured areas has a meaning, then use colours of similar luminance. Brighter colours tend to make areas look larger
- <http://hclwizard.org/>: palette creator and colour deficiency emulator
- Top R colour palettes
- All R colour palettes (including interactive colour picker)
- Colour palette finder
- PrettyCols

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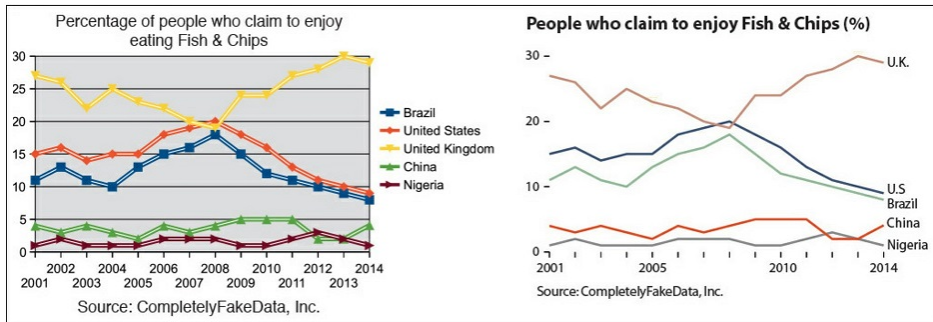
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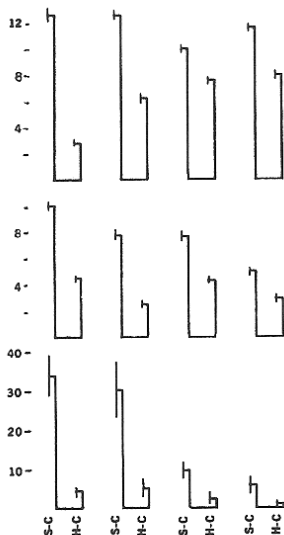
Aesthetics: Tufte

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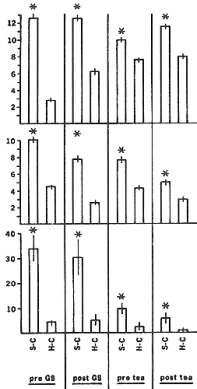
Which figure do you prefer?



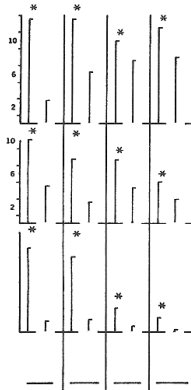
Noise removed



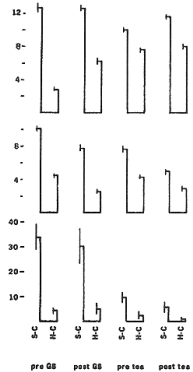
Hence



=



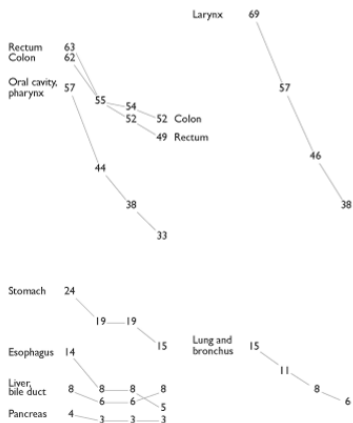
+



	Relative survival, % (SE)			
	5 years	10 years	15 years	20 years
Cancer site				
Oral cavity and pharynx	56.7 (1.3)	44.2 (1.4)	37.5 (1.6)	33.0 (1.8)
Oesophagus	14.2 (1.4)	7.9 (1.3)	7.7 (1.6)	5.4 (2.0)
Stomach	23.8 (1.3)	19.4 (1.4)	19.0 (1.7)	14.9 (1.9)
Colon	61.7 (0.8)	55.4 (1.0)	53.9 (1.2)	52.3 (1.6)
Rectum	62.6 (1.2)	55.2 (1.4)	51.8 (1.8)	49.2 (2.3)
Liver and intrahepatic bile duct	7.5 (1.1)	5.8 (1.2)	6.3 (1.5)	7.6 (2.0)
Pancreas	4.0 (0.5)	3.0 (0.5)	2.7 (0.6)	2.7 (0.8)
Larynx	68.8 (2.1)	56.7 (2.5)	45.8 (2.8)	37.8 (3.1)
Lung and bronchus	15.0 (0.4)	10.6 (0.4)	8.1 (0.4)	6.5 (0.4)
Melanomas	89.0 (0.8)	86.7 (1.1)	83.5 (1.5)	82.8 (1.9)
Breast	86.4 (0.4)	78.3 (0.6)	71.3 (0.7)	65.0 (1.0)
Cervix uteri	70.5 (1.6)	64.1 (1.8)	62.8 (2.1)	60.0 (2.4)
Corpus uteri and uterus, NOS	84.3 (1.0)	83.2 (1.3)	80.8 (1.7)	79.2 (2.0)
Ovary	55.0 (1.3)	49.3 (1.6)	49.9 (1.9)	49.6 (2.4)
Prostate	98.8 (0.4)	95.2 (0.9)	87.1 (1.7)	81.1 (3.0)
Testis	94.7 (1.1)	94.0 (1.3)	91.1 (1.8)	88.2 (2.3)
Urinary bladder	82.1 (1.0)	76.2 (1.4)	70.3 (1.9)	67.9 (2.4)
Kidney and renal pelvis	61.8 (1.3)	54.4 (1.6)	49.8 (2.0)	47.3 (2.6)
Brain and other nervous system	32.0 (1.4)	29.2 (1.5)	27.6 (1.6)	26.1 (1.9)
Thyroid	96.0 (0.8)	95.8 (1.2)	94.0 (1.6)	95.4 (2.1)
Hodgkin's disease	85.1 (1.7)	79.8 (2.0)	73.8 (2.4)	67.1 (2.8)
Non-Hodgkin lymphomas	57.8 (1.0)	46.3 (1.2)	38.3 (1.4)	34.3 (1.7)
Multiple myeloma	29.5 (1.6)	12.7 (1.5)	7.0 (1.3)	4.8 (1.5)
Leukaemias	42.5 (1.2)	32.4 (1.3)	29.7 (1.5)	26.2 (1.7)

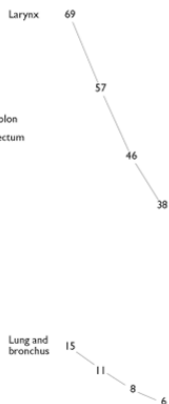
Digestive

Years 5 10 15 20



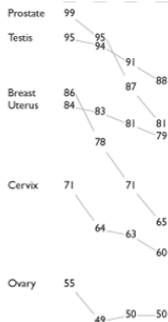
Respiratory

5 10 15 20



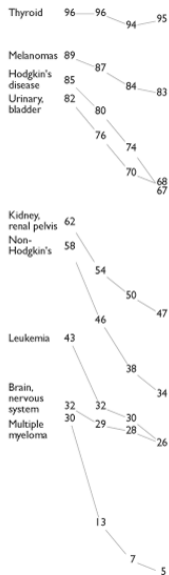
Reproductive

5 10 15 20



Other

5 10 15 20



Outline

Introduction

Principles of Data Visualisation

Scale and range

Theories of perception

Decoding: William Cleveland

A hierarchy of visual decoding

Size (numeric)

Group (categorical)

Colour

Aesthetics: Tufte

Final words

Main Message

- Last two decades: new ideas on what is the most informative way to display quantitative information
 - Based on scientific knowledge on perception
 - Guidelines that help improving graph quality. “Mistakes” due to ignorance
 - No “right or wrong”, only “better or worse”
- Graph creation iterative and laborious process, with trial and error.

Creating a half-page figure takes same time as writing half-page text
- Several high quality newspapers and magazines (New York Times, Financial Times, Washington Post, The Economist) are ahead of many scientists and institutions (including WHO)

Graphics principles, the STRATOS initiative

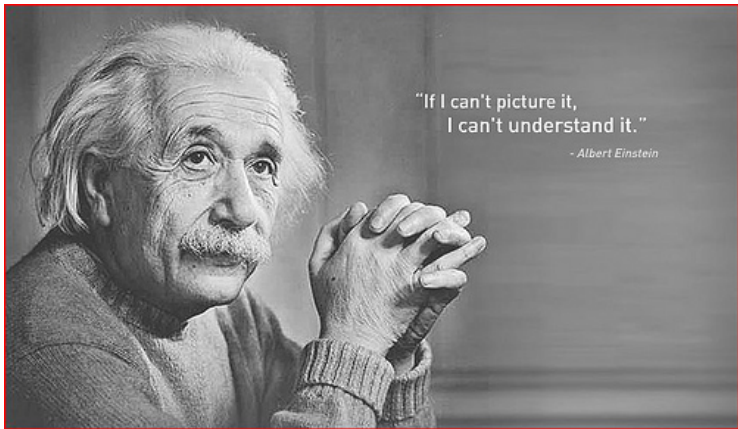


From [https:](https://)

[//graphicsprinciples.github.io/threelaws.html](https://graphicsprinciples.github.io/threelaws.html)

STRATOS

(STRengthening Analytical Thinking for Observational Studies)
visualisation panel



"If I can't picture it,
I can't understand it."

- Albert Einstein

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